

Objectives

1. the students will be able to appreciate the historical background/perspective, from Hippocrates, to A.T. Still, to consideration of the 5 (or 6) Osteopathic Models, to Dr. Phil Greenman's legacy, and on to the related to inter-professional and global awareness of the Lower (and Upper) Crossed syndrome, from the legacy of neurologists, Drs. Janda and Lewit.
2. the students will be able to understand the Lower Crossed Syndrome, including the role of the type I (slow twitch, postural) and type II (fast twitch, phasic) muscles. The student will understand which muscles need stretching, and which muscles need strengthening, in relation to application of an understanding of this "crossed" syndrome. Specific exercises will be reviewed at the end of the lecture, with pictures and diagrams included.
3. the students will be able to understand the importance of obtaining and considering the patient's history (subjective), along with objective information. This objective information can include Special Tests, muscle testing/palpation, the osteopathic structural examination, postural observation and analysis, and other neurological (NCV / EMG) or radiological (x-rays, MRI, etc.) testing, as needed.
4. the students will be able to understand how to incorporate an understanding of postural compensation, tensegrity (a.k.a. biotensegrity), gravitational strain pathophysiology (G.S.P.) and postural decompensation into their osteopathic perspective.
5. the students will read the assigned article from Kim (LINK: <https://pubmed-ncbi-nlm-nih-gov.nyit.idm.oclc.org/30899747/>), which is also listed under the Resources for this Classroom Session) on their own, to gain a sense of the clinical correlation / clinical suggestion related to this topic area.

Outline

- Historical perspective
- Review Muscle Types (I and II)
- Lower (Pelvic) Crossed Syndrome
- Other Postural Considerations
 - Postural Compensation
 - Tensegrity
 - Gravitational Strain Pathophysiology (G.S.P.) and Postural Decompensation
- General Treatment Strategies

Historical Perspective

Vladimir Janda, M.D., D.Sc. (1928-2002)

- H/o poliomyelitis as a teenager, wheelchair-bound for 3 years after an initial quadriplegia.
- **Neurologist** training at Charles University, Prague.
- 1960s in Canada, he did post-doctoral work with Professor John Basmajian.
- His work in rehabilitation included research and clinical activities, over many decades and with a global influence.
- He was the Editor in Chief of the Czech Journal, Rehabilitation and Physical Medicine, until 2002.
- He worked with **Karel Lewit, M.D.**, another Czech **neurologist**, **presenting the Upper & Lower Cross Syndromes.**



Vladimir Janda, MD, DSc 1928–2002.

In clinical practice, we distinguish two muscle types in the evaluation and treatment:

**Type I muscles = postural muscles =
slow twitch muscles**

These muscles react by becoming TIGHT or SHORTENED as they become “facilitated” in response to postural imbalance, injury, somatic dysfunction.

Thus it is good to know which muscles are being categorized as postural/slow twitch muscles.

**Type II muscles = phasic muscles =
fast twitch muscles**

These muscles react by becoming WEAK as they become “inhibited,” For instance, the Gluteus medius muscle, a phasic muscle, may become reflexly inhibited due to an associated sacroiliac joint issue (e.g. extended, torsion, etc.) on one side and thus causing the muscle to become relatively weak on that side vs. the other side. The muscle itself has “not lost its innate strength” but rather is inhibited and thus becomes weak as a result.

Type I and Type II muscles in the clinical setting

Fiber Type	Postural or Phasic	Slow twitch or Fast twitch	Clinically – when involved – this muscle type tends to:	Examples in the clinical setting
TYPE I	Postural	Slow twitch	Become facilitated and thus SHORTENED or TIGHT	Upper crossed syndrome: Example 1: Head forward posture – <u>Anterior tight/shortened</u>= facilitated muscles: Pectorales <u>Posterior tight /shortened</u>=facilitated muscles: Levator scapulae, upper trapezius, SCM Example 2: when there is excessive ankle pronation – one can see the lower crossed syndrome – <u>Anterior: tight</u> iliopsoas muscle and <u>Posterior: tight</u> erector spinae muscles

Type I and Type II muscles in the clinical setting

Fiber Type	Postural or Phasic	Slow twitch or Fast twitch	Clinically – when involved – this muscle type tends to:	Examples in the clinical setting
Type II	Phasic	Fast twitch	Become <u>inhibited</u> and thus WEAK	Upper crossed syndrome (example: head forward posture) <u>Anterior: weak</u> neck flexors=inhibited <u>Posterior: weak</u> rhomboids/serratus anterior LOWER crossed syndrome: <u>Anterior: weak/inhibited</u> abdominals <u>Posterior: weak/inhibited</u> gluteus maximus/medius

Upper Crossed and Lower Crossed Syndromes

Clinically (e.g. work by Vladimir Janda, M.D./Prague/Czech Republic) there have been observed PATTERNS of muscle involvement, that have been called the **upper and lower crossed syndromes**. “Upper” because it involves the upper body, and “lower” because it involves the pelvis, or lower body.

It is called “Crossed” as there are two ‘lines’ that form a cross-over with ANTERIOR muscles being facilitated=tight=shortened (type I muscles) and POSTERIORLY (follow one LINE – call it the “facilitated” or “shortened” line) with type I muscles being also shortened. This is in response to a muscle imbalance.

Similarly, follow the other cross-line that goes from ANTERIOR to POSTERIOR, representing the INHIBITED and thus WEAK muscles both anteriorly and posteriorly.

So it is like a “puzzle” where you have two lines with four “stations:” 2 anterior and 2 posterior groups. Knowing the four “groups” – will give you an understanding of which muscles can be involved at any one time – so if the pectoralis major is tight in front, and you do have a cross syndrome – upper, for instance – you can deduce which other muscles are involved in a particular patient.

Table 1 – Anterior Muscles involved in Upper and Lower Crossed Syndromes

memorize

Anterior Postural Muscles	Anterior Phasic Muscles
Upper Cross	
Scalenes	Deep cervical flexors
Sternocleidomastoid	
Pectoralis major (upper)	
Pectoralis minor	
Lower Cross	
Iliopsoas	Rectus abdomini
Rectus femorus	Three Vasti muscles
Adductor brevis	Tibialis anterior

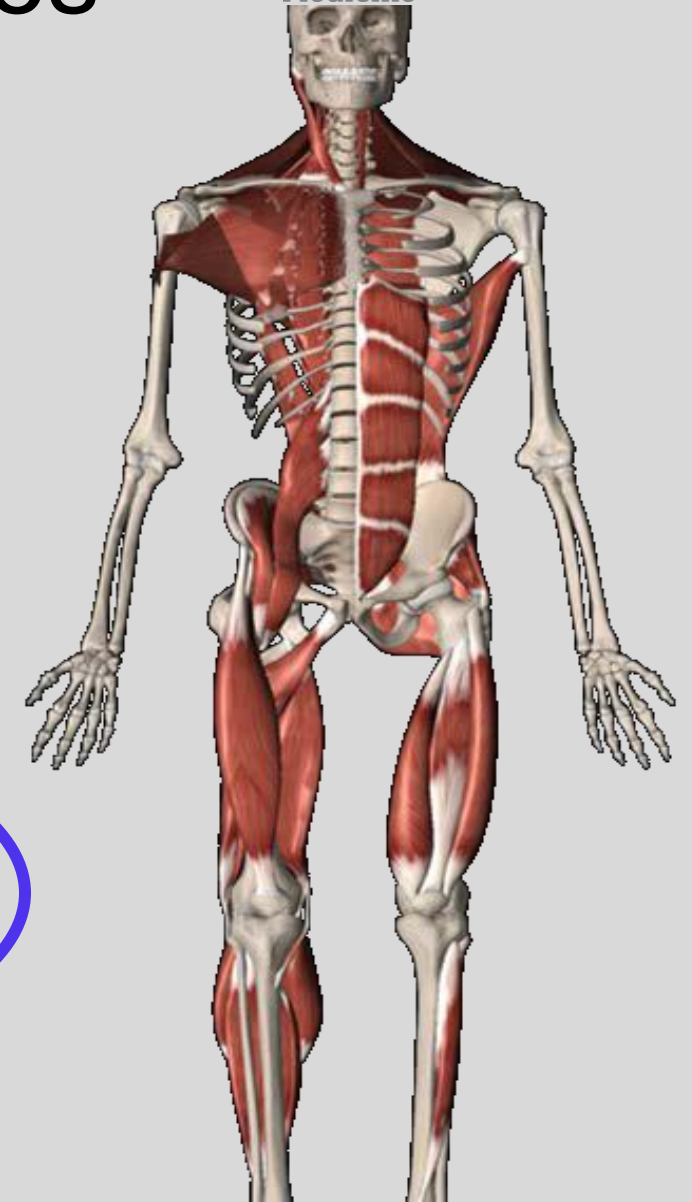
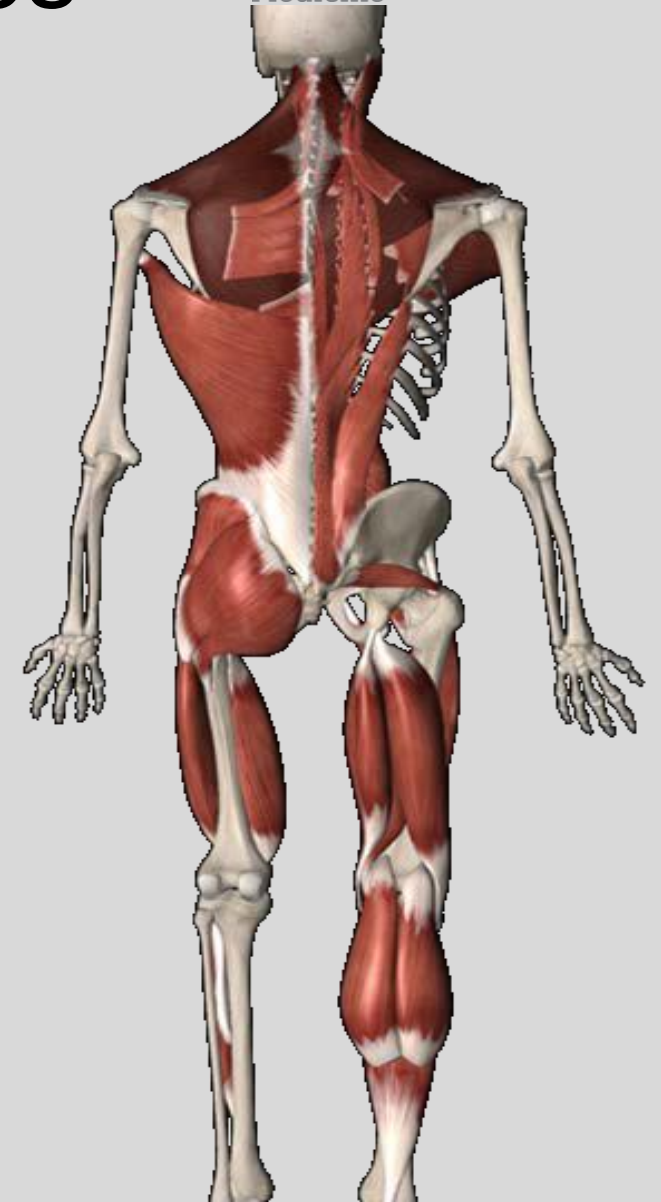


Table 2 – Posterior Muscles involved in Upper and Lower Crossed Syndromes

memorize

Posterior Postural Muscles	Posterior Phasic Muscles
Upper Cross	
Upper Trapezius	Mid-lower Trapezius
Levator Scapulae	Rhomboids
Cervical erector spinae	Latissimus dorsi
Lower Cross	
Quadratus lumborum	Gluteus minimus
Thoracolumbar erector Spinae	Gluteus medius
Hamstrings	Gluteus maximus
Piriformis	
Gastroc-soleus complex	

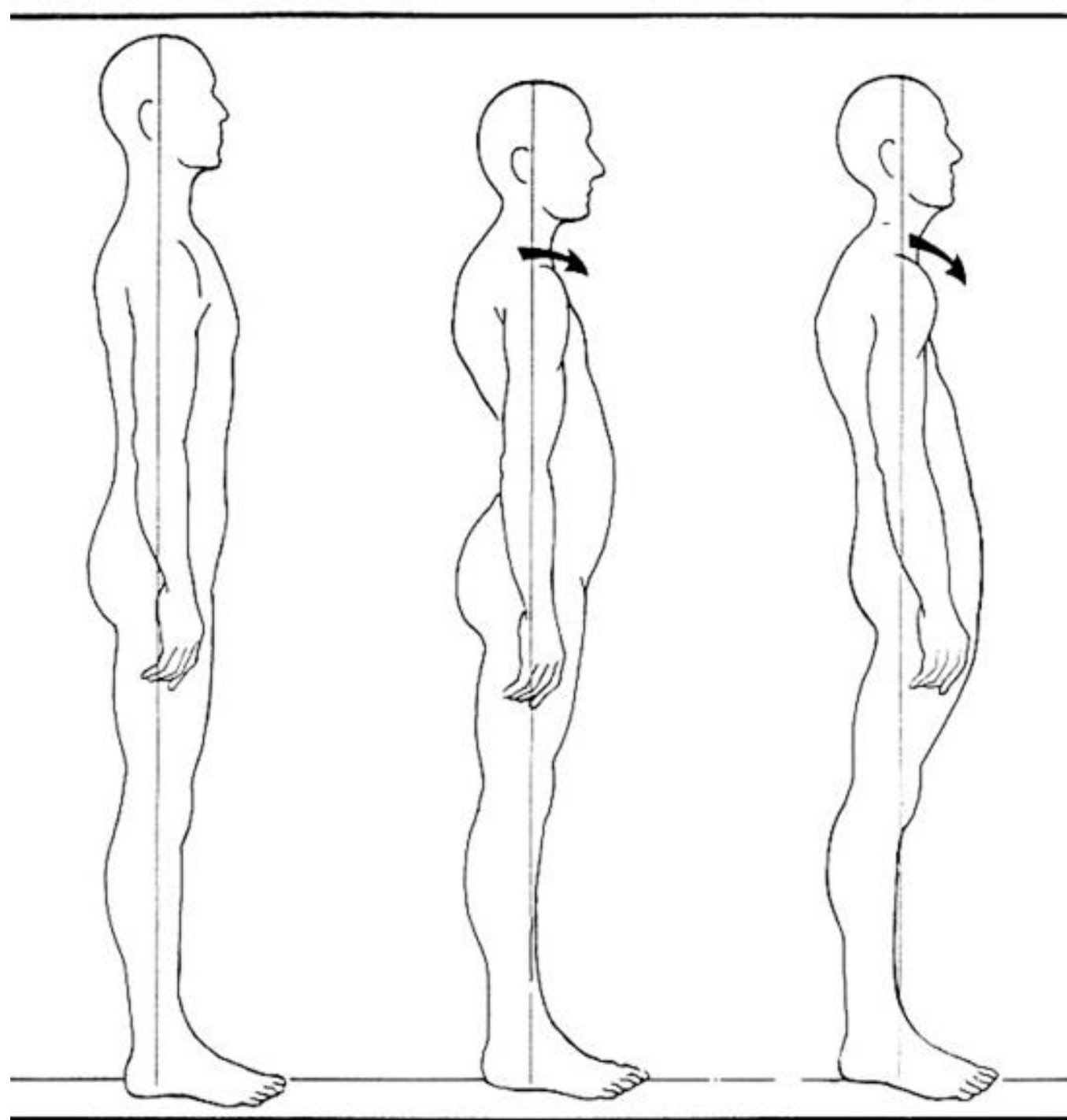


Clinical Association: PRONATION OF THE SUBTALAR JOINT and Lower Crossed Syndrome

- PRONATION of the subtalar joint and LOWER CROSSED SYNDROME
- Mechanical stress at the knee (genu valgus/valgum), hip and lumbar spine
- **Muscle imbalance** - tight/facilitated type I muscles vs. weak/inhibited type II muscles – KEY muscle (often) → GLUTEUS medius weakness

- Symptoms-compensatory or facilitative mechanisms may cause:

- Cervical, thoracic, lumbosacral pain
- GI disorders, abdominal, pelvic and inguinal pain
- Headache, neck pain



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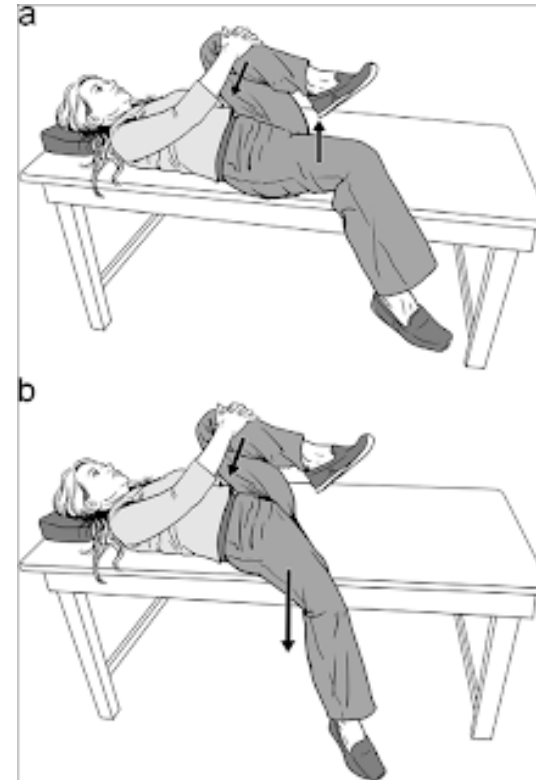
Iliopsoas Hip Flexor Stretch - Kneeling



Kneel and place opposite foot significantly forward
Hold on to couch or bed and move your pelvis forward while keeping your shoulders back
Feel stretch in front of hip of leg that is on the ground
Hold 30-60 seconds

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Iliopsoas Hip Flexor Stretch – on back



Lie on bed on back. Pull one knee up towards chest.
Other foot stays hanging off the bench.
Hold for 30-60 seconds. Repeat on other side.
Feel stretch in front of hip of leg that is hanging.

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Hamstring Stretch



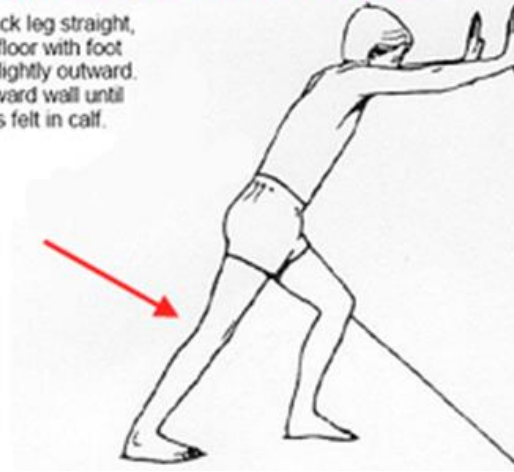
Sit on edge of solid chair
One leg is straight, the
other bent
Reach toward knee and
beyond if possible of
straight leg
Feel stretch in back of
straight leg
Hold 45 seconds

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Gastroc-soleus complex stretch

Straight Leg Stretch for Gastroc

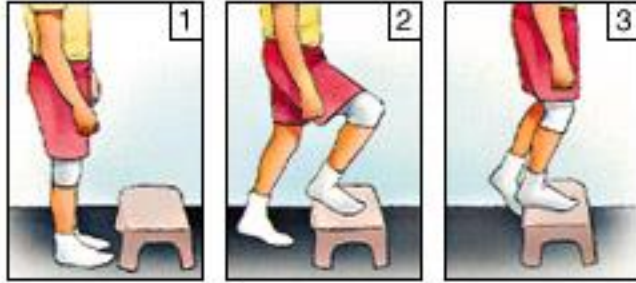
Keep back leg straight,
heel on floor with foot
turned slightly outward.
Lean toward wall until
stretch is felt in calf.



Hold on to solid object
such as kitchen
counter/wall
Place balls of feet on
2x4 or something a
little higher
Make sure heels are
on ground and knees
are straight
Feel stretch in calf
Hold 1-2 minutes

Vastus Intermedius, Vastus Lateralis, Vastus Medialis

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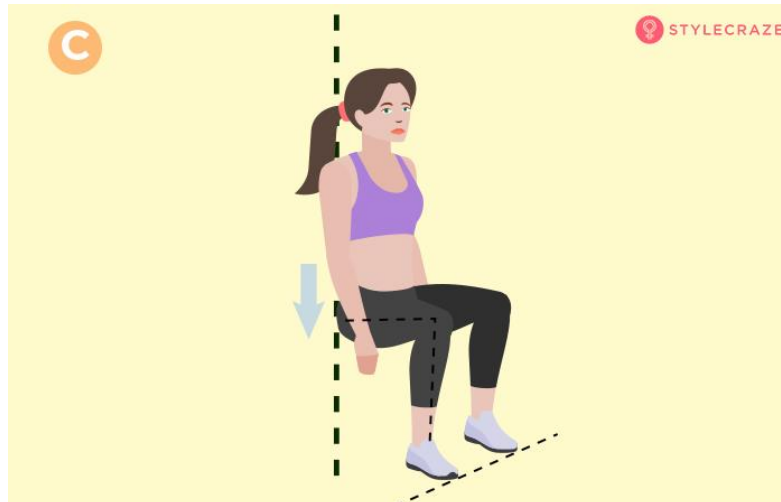


Step up onto a bench straightening the knee that you step up with. Repeat 8-15x.

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Stand up on a step. **Hold on to something for balance.** Lower a heel onto the floor but do not put weight on it. Just touch and come back up to starting position. Repeat 8-15 times.



Keep back flat against wall, slide down until legs are at 90 degrees. Shift weight to the heels



Summary Slide

- The Lower (Pelvic) Crossed Syndrome provides a clear framework from which to view a patient's postural compensations and subsequently target treatment approaches (*which can include an Exercise Prescription, as noted in the examples*).
- Other postural considerations are manifold and important to consider in a patient's clinical presentation.
- Application of the osteopathic perspective and the osteopathic philosophy can be beneficial for a patient's overall well being.